SHAH ABDUL MAZID

AI/ML ENGINEER

Generative AI | LLMs | RAG | Computer Vision | NLP

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Professional Summary

AI/ML Engineer with a B.Sc. in Computer Science and Engineering from East West University, majoring in Intelligent Systems and Data Science. Experienced in Generative AI, LLMs, RAG, Machine Learning, Deep Learning, Computer Vision, and NLP. Skilled in building AI applications using Python, PyTorch, TensorFlow, FastAPI, Pinecone, OpenAI APIs, and AWS. Passionate about AI research and developing intelligent solutions for real-world problems.

Technical Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL

AI / ML: Machine Learning, Deep Learning, Transfer Learning, Model Evaluation, Data Preprocessing

Generative AI: Generative AI, LLMs, RAG, Prompt Engineering, AI Agents, Agentic AI, Multi-Agent Systems

NLP: Natural Language Processing, Semantic Search, Vector Search, Embeddings, Sentence Transformers, Document Q&A, AI Chatbots

Computer Vision: Object Detection, Image Classification, Medical Imaging, YOLO, OpenCV, Grad-CAM, CBAM Attention

Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, Plotly

Backend: FastAPI, REST APIs, JWT Authentication, API Integration, PDF Processing

Frontend: React, Next.js, Streamlit, Gradio

Databases: Pinecone, Vector Databases, MongoDB, MySQL, SQLite

Cloud / DevOps: AWS, EC2, Docker, PM2, Git, GitHub

Automation: n8n, Workflow Automation, API Integration

Education

B.Sc. in Computer Science & Engineering **2021–2026**
East West University
Major: Intelligent Systems & Data Science
Result: 2.56

Higher Secondary Certificate **2018–2020**
Dhaka Ideal College
Group: Science
Result: 4.08

Secondary School Certificate **2016–2018**
Badshah Faisal Institute
Group: Science
Result: 4.00

Languages

Bengali — Native

English — Professional Working Proficiency

Research Interests

- Generative AI and Large Language Models
- Retrieval-Augmented Generation
- Multi-Agent AI Systems
- Healthcare AI
- Computer Vision and Medical Imaging
- Natural Language Processing
- Intelligent Recommendation Systems
- Model Interpretability

Career Focus

AI/ML Engineering

Generative AI

LLM and RAG Systems

AI Research

Computer Vision

Data Science

Tools & Platforms

Models & APIs: OpenAI API, Llama 2, Mistral, Claude API, HuggingFace Models

Vector DBs & Search: Pinecone, Weaviate, Milvus, Chroma, FAISS

Deployment: Hugging Face Hub, AWS SageMaker, Docker Hub, Streamlit Cloud

Dev Tools: VS Code, Git, GitHub, Jupyter, Google Colab, Linux/Ubuntu

APIs & Integrations: OpenAI API, Google Maps, Stripe, Google Sheets, Anthropic API

Key Achievements

- Published 2 research papers in agricultural AI and medical imaging (ICCIT 2025, Springer ICIDA 2026)
- Built enterprise RAG platform processing 10,000+ documents with 95%+ accuracy
- Developed traffic detection system for real-world Bangladesh traffic conditions
- Designed multi-agent AI system handling complex enterprise workflows
- 17+ IBM and AWS certifications in AI/ML specializations

Professional Highlights

Published Research: Published Research: 2 publications in agricultural AI and medical imaging, including IEEE ICCIT 2025 and Springer ICIDA 2026. services.

Generative AI & RAG: Built enterprise document Q&A, retrieval-augmented, and multi-agent AI systems.

End-to-End AI: Developed ML systems spanning model development, backend integration, and production deployment.

AI Automation: Built n8n-based workflows integrating AI models, APIs, and external

Professional Experience

AI Engineer
July 2026 – Current

Neuroxyte — Narayanganj, Dhaka Division, Bangladesh

- Develop and deploy AI solutions using Generative AI, LLMs, RAG, and intelligent automation.
- Build AI applications and scalable backend services using Python and FastAPI.
- Design LLM-based workflows, conversational AI systems, and retrieval-augmented applications.
- Integrate AI models, external APIs, and third-party services to deliver reliable end-to-end solutions.

AI Engineer
Feb 2026 – July 2026

Softvence Agency — Dhaka, Bangladesh

- Developed RAG-based chatbots using Large Language Models for intelligent question answering.
- Built FastAPI backend services and integrated AI models and external APIs.
- Developed Text-to-Speech and voice generation systems for interactive AI applications.
- Automated business workflows using n8n and implemented Generative AI and RAG solutions.

Selected Projects

Nexus Intelligence [GitHub] [Live Demo]

Enterprise Multi-Agent AI Platform

Built a full-stack enterprise AI platform for secure question answering across private documents, spreadsheets, and web data.

- Implemented RAG, multi-agent routing, Pinecone retrieval, feedback memory, JWT authentication, and PII detection.
- **Tech:** FastAPI, Next.js, React, TypeScript, OpenAI, Pinecone, AWS.

HR Policies RAG Chatbot [GitHub]

AI-Powered Document Q&A

Built a citation-aware RAG chatbot for answering HR policy questions from organizational PDF documents.

- Implemented PDF ingestion, semantic search, vector retrieval, and LLM-based response generation with source-aware answers.
- **Tech:** Python, FastAPI, Streamlit, Pinecone, Llama 2, Sentence Transformers.

Bangladesh Traffic Flow Dataset : [GitHub]

Deep Learning Vehicle Detection

Developed a computer vision system for vehicle detection and classification under real-world Bangladesh traffic conditions.

- *Implemented YOLO-based detection with Grad-CAM/EigenCAM interpretability and interactive inference workflows.*
- ***Tech:** PyTorch, YOLO, OpenCV, NumPy, Pandas, Streamlit, Gradio.*

Brain Tumor Detection and Classification : [GitHub]

MobDenseNet with CBAM Attention

Developed a hybrid MobileNetV2–DenseNet121 architecture with CBAM attention for MRI-based brain tumor classification.

- *Applied patient-level data splitting, augmentation, classification, and Grad-CAM for interpretable predictions.*
- ***Tech:** PyTorch, MobileNetV2, DenseNet121, CBAM, Grad-CAM, Streamlit.*

WhatsUpIn : [GitHub]

AI-Powered Travel Recommendation Engine

Built an AI-powered travel platform that generates personalized itineraries and automates travel-related workflows.

- *Integrated OpenAI, n8n, Google Maps, Google Sheets, and Stripe to automate itinerary generation and travel operations.*
- ***Tech:** OpenAI, n8n, Google Maps, Google Sheets, Stripe, REST APIs.*

Publication

MangoStack: A Lightweight, Interpretable Ensemble for Real-Time Mango Leaf Disease Diagnosis

***Authors:** Raiyan Gani, Yusuf Salehin, Md. Shakil Bhuiyan, **Shah Abdul Mazid**, Monisha Bani Nibedita Shuci, Shamim Ripon*

2025 28th International Conference on Computer and Information Technology (ICCIT), IEEE, pp. 2235–2240.
DOI: 10.1109/ICCIT68739.2025.11490181

Uncertainty-Aware Chest Pathology Detection with Vision and Language Transformers

***Authors: Shah Abdul Mazid**, Monisha Bani Nibedita Shuci, Mahia Mehrun Safa, Md. Ashikur Rahman Anik, Md. Omor Faruk Sejan, Khandhakar Shatu Moni, Md. Sazzad Hossain, Md. Adnan Morshed, Ahmed Wasif Reza*

Innovations in Data Analytics, Springer Nature, 2026, pp. 397–417.
DOI: 10.1007/978-3-032-27845-6_34

Certifications

- *IBM Deep Learning with PyTorch, Keras and TensorFlow*
- *DeepLearning.AI TensorFlow Developer*
- *IBM Machine Learning*
- *IBM RAG and Agentic AI*
- *Building AI Agents and Agentic Workflows*
- *IBM AI Developer*
- *AWS Generative AI and AI Agents with Amazon Bedrock*
- *IBM AI Engineering*
- *IBM Data Science*

Competitions & Awards

***Network War** — EWU Telecommunication Club, 2024*

***IT Olympiad** — CSE FEST, East West University, 2024*

***In House Programming Battle** — EWUCoPC, 2022*

Online Profiles

***Portfolio:** Portfolio Profile*
***GitHub:** Github Profile*
***LinkedIn:** LinkedIn Profile*
***Nexus Intelligence:** Live Demo*